service in 1922. It was not an impressive vessel – a converted collier with a top speed of 14 knots – but it was a start. The next two were much larger and faster. USS *Lexington* and USS *Saratoga* were originally meant to be battle cruisers. But at the Washington Conference in 1922 the leading naval powers agreed to limits on warship numbers. The battle cruisers could not be built – but two fleet carriers could. *Saratoga* and *Lexington* were each capable of carrying up to 81 aircraft and had a top speed of 34 knots – faster than any warship of comparable size. In naval exercises from 1929 onward they proved their ability to play a key role as an offensive strike force. Notably, in 1932 more than 150 airplanes from the two carriers executed a mock attack on the naval base at Pearl Harbor, which achieved total surprise and would have had a devastating effect if really carried out by an enemy power – the Japanese, for example.

Naval airships

Carriers were not the only means of providing aerial support at sea that were explored by the United States. In the early 1930s, experiments were also conducted with the naval airships *Akron* and *Macon*. They were designed to carry fighter planes, which they would launch and recover in the air – the returning airplane had to adjust its speed to that of the airship, position itself below the airship's hull, and then fly upward to hook itself on to a support sticking out from the hull. These airborne aircraft carriers could accompany the fleet, acting as command and control centers for their airplanes, which would carry out reconnaissance missions and provide air defense. This bizarre-seeming idea might have worked but for the vulnerability of airships, not only to enemy action but also to the weather. The *Akron* was lost at sea in 1933 and the *Macon* suffered a similar fate in 1935. The only lighter-than-air aircraft the US Navy continued to use were nonrigid blimps.

Carrier fleets

Despite financial stringencies, the US Navy achieved a respectable development of its carrier force in the 1930s, with the addition of the USS *Yorktown* and *Enterprise* thanks to money from Roosevelt's New Deal programs. But the exact role of carriers remained undecided, with many senior commanders still convinced that traditional warships held the key to sea victory. The Japanese, already identified as America's most likely enemy in a naval conflict, also prevaricated about the role of carrier aviation. But they accepted that aircraft would have to be used to weaken the US fleet before Japan's heavy warships could deliver a knockout blow. On this basis the Japanese developed the world's most effective carrier force in the 1930s, with better aircraft and better-trained aircrews than their American counterparts.

AIRCRAFT FOR CARRIERS

During the interwar years, US naval air tacticians worked out a basic mix of aircraft for carriers. The three main types were fighters, for the defense of the fleet and to achieve air superiority; dive-bombers to attack enemy ships from above; and low-flying torpedo aircraft.

On the whole, naval air forces lagged behind land-based forces in making the transition from slower biplanes to higher performance monoplanes. The Royal Navy was especially archaic in introducing the Fairey Swordfish biplane – top speed 138mph (222kph) – as its latest torpedo bomber in 1934, but the US Navy still had the Boeing F4B biplane as its main fighter through most of the 1930s. There were some good reasons for sticking to biplanes: they could land at lower speeds, a useful attribute at sea, and tended to take up less deck space than monoplanes (which needed a greater wingspan to achieve the same lift). However their slow speeds made them vulnerable to antiaircraft fire during their long approaches on target when delivering their torpedoes.

LEGENDARY "STRINGBAG"
Introduced in 1934, the Fairey Swordfish was a three-man torpedo bomber that could also be used for antisubmarine warfare. Remarkably, this open-cockpit biplane performed admirably in World War II.

BIPLANE LANDING
Sailors standing in the USS Langley*'s safety nets watch as a US Navy Aeromarine 39-B biplane successfully lands on the carrier's deck in October 1922.*